Understanding Inflation

Why Prices Rise and How Central Banks Respond

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Three questions for today

1. What is inflation and how is it measured?
2. Is inflation good or bad?
3. What do central banks do, and at what cost?

A receipt from the same store, 24 years apart



Item	2002	2026	Change
Rice, 5 kg	¥980	¥4,200	+329%
Milk, 1 L	¥98	¥260	+165%
Eggs, 10	¥78	¥310	+297%
Bread, 1 loaf	¥68	¥210	+209%
Cabbage	¥58	¥230	+297%
Total	**¥1,282**	**¥5,210**	**+306%**

Store: JUSCO Hadano, 12 May 2002 → AEON Hadano, 12 Oct 2026

(same supermarket; JUSCO was later renamed AEON)

Average of the five product increases: $(329 + 165 + 297 + 209 + 297) \,/\, 5 = \mathbf{259\%}$

Average inflation per year (compound): $\left(\dfrac{\text{final}}{\text{initial}}\right)^{1/\text{years}} - 1 = \left(\dfrac{5210}{1282}\right)^{1/24} - 1 \approx \mathbf{6\%}$

Do you think 6% is the inflation rate?

Milk rose 165%; rice rose 329%.
Whose basket — and how should items be weighted?

The Consumer Price Index

1. Definition

The cost of a fixed basket of goods and services, relative to a base period.

Basket from household spending surveys

(average purchases in the base year).

2. Weights

Weight = category spending ÷ total spending.

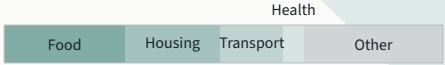
Larger share of spending ⇒ larger weight.

3. Formula

$$\mathrm{CPI}_t = 100 \times \frac{\sum_j p_{tj}\, q_j}{\sum_j p_{0j}\, q_j}$$

j = item · p_{tj} = today's price · q_j = basket quantity

What's in the basket



Food	27.5%
Housing	21.8%
Transport	14.4%
Health	4.7%
Other	31.5%

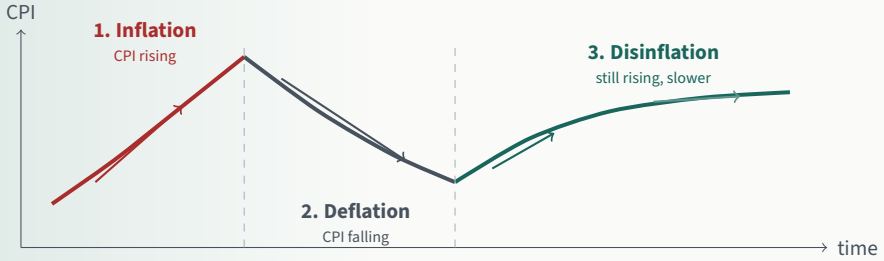
Statistics Bureau of Japan, 2025 base

Inflation, Deflation, and Disinflation

CPI = a *level*. The inflation rate measures how fast that level is changing:

$$\pi_t = \frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}}$$

1. Inflation: CPI rising ($\pi > 0$). **2. Deflation:** CPI falling ($\pi < 0$). **3. Disinflation:** CPI still rising, but π falling.



Example: milk and rice

Question

Basket = **2 L milk** + **2 bags rice**. Compute the CPI at $t = 0, 1, 2$, then π_1 and π_2. Is $t = 2$ deflation?

Prices

	$t = 0$ (base)	$t = 1$	$t = 2$
Milk	¥100	¥120	¥125
Rice	¥200	¥240	¥250

Example: milk and rice

Question

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Solution

	$t = 0$	$t = 1$	$t = 2$
Cost $= 2p_m + 2p_r$	$2 \times 100 + 2 \times 200 =$ **600**	$2 \times 120 + 2 \times 240 =$ **720**	$2 \times 125 + 2 \times 250 =$ **750**
CPI $= 100 \times$ cost$/600$	**100**	**120**	**125**
π_t	—	$(120 - 100)/100 =$ **20%**	$(125 - 120)/120 \approx$ **4.2%**

CPI rose both periods $\Rightarrow$ **inflation**. π fell 20% $\to$ 4.2% $\Rightarrow$ **disinflation**, not deflation.

Three costs of inflation

1. Shoe-leather

Cash loses value when prices rise. People hold less cash and make more trips to the bank — time wasted protecting money.

2. Menu costs

Firms change prices more often (menus, websites, shelves). Prices update at different speeds $\Rightarrow$ price confusion.

3. Unexpected inflation

Moves wealth around when inflation differs from expectations: borrowers gain, lenders and fixed pensions lose. Unpredictable inflation creates uncertainty — planning gets harder for everyone.

Benefits of inflation

1. Encourage investment

Modest inflation makes idle cash lose a little value, nudging people toward productive investment (stocks, equipment, new firms).

2. Grease the job market

Firms rarely cut nominal pay. With a little inflation, real wages can adjust down without a pay cut — helping keep unemployment lower.

Hyperinflation: Zimbabwe

What it is	Extremely rapid price increases — prices can double in hours or days.
Zimbabwe	2008 peak: inflation exploded; people abandoned the Zimbabwean dollar, turned to foreign currency and barter.
Why	Money printed to cover deficits; confidence in the currency collapsed.
Today	Argentina and Venezuela still hyperinflationary; Zimbabwe's new currency is still watched closely.

Tool 1: Interest rates

Which tool to use depends on how costly inflation is.

Raise rates	Borrowing gets more expensive $\Rightarrow$ spending and investment cool $\Rightarrow$ demand falls $\Rightarrow$ inflation eases.
Lower rates	Borrowing gets cheaper $\Rightarrow$ spending and investment rise $\Rightarrow$ demand rises $\Rightarrow$ can push inflation up.
Tradeoff	Fighting inflation with higher rates often means slower growth.

Tool 2: Reserve requirements

The share of deposits banks must keep as reserves.

Raise the ratio	Banks must hold more cash idle ⇒ less lending ⇒ money and demand tighten ⇒ inflation pressure falls.
Lower the ratio	Banks can lend more ⇒ credit expands ⇒ spending can rise ⇒ inflation pressure rises.
Idea	Control credit at the source — how much banks are allowed to create.

Tool 3: Yield curve control

Japan's case — pinning long-term interest rates.

What it is	The central bank targets not only short rates, but also long-term bond yields.
Bank of Japan	Bought government bonds as needed to keep long yields near a target — cheap long-term borrowing.
Why	When short rates are already near zero, YCC tries to keep stimulus flowing through the whole yield curve.

Diagnosing the cause

Three channels — illustrations from the United States.

1. Demand	COVID “helicopter money”: cash sent to households $\Rightarrow$ spending rises faster than supply $\Rightarrow$ prices are pulled up.
2. Supply	Oil shocks (Gulf / Hormuz) and trade wars: higher energy costs, tariffs, broken chains $\Rightarrow$ costs rise $\Rightarrow$ prices are pushed up.
3. Expectations	Inflation high $\Rightarrow$ workers expect higher prices $\Rightarrow$ demand higher wages $\Rightarrow$ firms pass on costs $\Rightarrow$ inflation stays high $\Rightarrow \ldots$

The output–inflation tradeoff

Different causes need different responses.
Fighting a supply shock only by cutting demand can cause **stagflation**.

Demand-driven

Raise rates ⇒ spending cools ⇒ inflation falls.

Cost: slower growth.

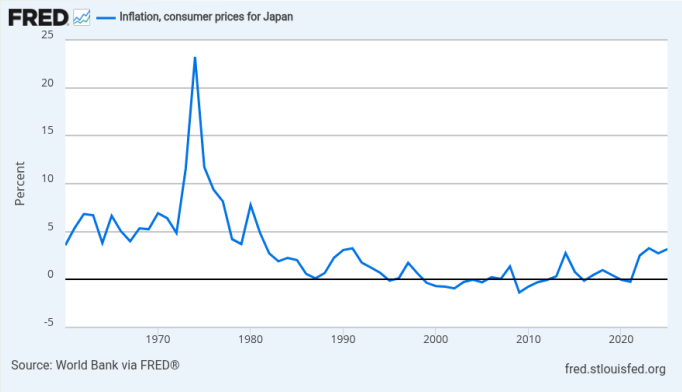
Supply-driven

Oil / trade war: prices up, output down — **stagflation**.

Cutting demand alone deepens the slump.

Case study: Japan’s CPI inflation over time

Annual % change in consumer prices. Source: World Bank via FRED (series FPCPITOTLZGJPN).



Periods to read: bubble peak ∼1990 → long near-zero/negative stretch → 2008–09 dip → 2013–14 uptick → COVID dip → 2022+ pickup.

Reading the chart: bubble, deflation, 2008

1. Bubble ∼1990	Late-1980s stock and land boom; BOJ raised rates in 1989–90; equity prices collapsed in 1990. CPI inflation was still about 3% in 1990–91, then faded as the bust hit balance sheets.
2. Long deflation	CPI inflation turned negative in 1995 and stayed near zero or below for years: weak demand, bad loans, and expectations that prices would keep falling. BOJ response: zero rates (1999), QE (2001), then QQE / negative rates / YCC (2013–16).
3. 2008–09 dip	Inflation briefly positive in 2008, then about -1.4% in 2009 — oil prices crashed after the 2008 surge, and demand weakened in the global crisis.

Reading the chart: Abenomics, COVID, turnaround

4. 2013–14 uptick

April 2013: QQE and a 2% price target. 2014 CPI rose about 2.7%, but much of that was the consumption-tax hike (5%→8%); underlying inflation stayed softer.

5. COVID dip

All-items CPI: 0.0% in 2020, −0.2% in 2021 — prices slipped again during the pandemic lull.

6. 2022+ pickup

Core CPI (excl. fresh food) above 2% since April 2022. Drivers: weaker yen, imported energy costs, then stronger wages (2024 spring talks: largest in decades).

March 2024: BOJ ended negative rates and YCC — from fighting deflation to managing inflation.